

Coding & Solution

Date	7 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Crop Recommendation System
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	GitHub: https://github.com/Bhavagna2007/OptiCrop Live Demo: https://opticrop-wii6.onrender.com/
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	• Crop recommendation using Machine Learning • Predicts crop based on Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall • User-friendly web interface • Displays recommended crop with description • Input validation and error handling • Responsive design • Deployed on Render
Pending / Incomplete Features	• User authentication • Prediction history storage • Weather API integration • Fertilizer recommendation module
Setup / Run Instructions	1. Clone the GitHub repository. 2. Install dependencies using <code>pip install -r requirements.txt</code>.

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| | <ol style="list-style-type: none">3. Run the application using <code>python app.py</code>.4. Open <code>http://localhost:5000</code> in a web browser. |
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Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop application is a machine learning-based crop recommendation system developed using Python, Flask, and Scikit-learn. The application accepts soil nutrient and environmental parameters as input and predicts the most suitable crop. It has been successfully deployed on Render and integrated with GitHub for version control and continuous deployment.